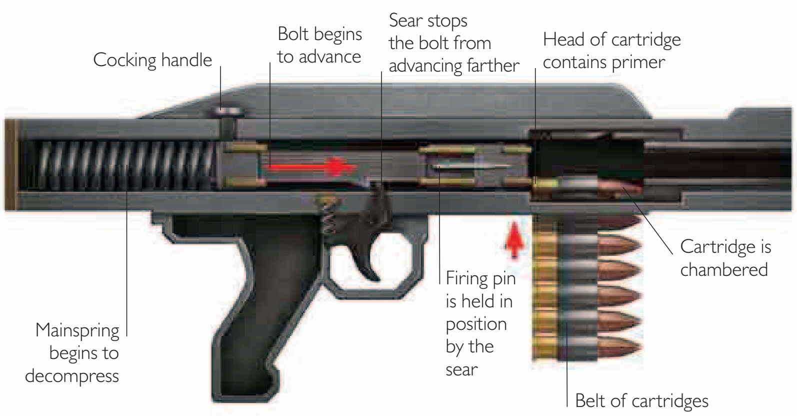
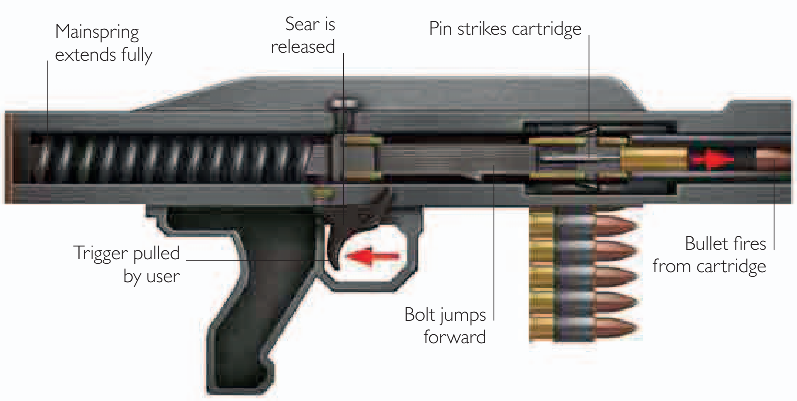
## Recoil reloading

Every action, Isaac Newton's Third Law of Motion tells us, has an equal and opposite reaction. The action—ignition of the propellant—in a firearm—propels the bullet down the barrel and on toward its target. The reaction, known as the recoil, drives the gun into the shoulder or hand of the user. Recoil-operated action drives the auto-loading action of many semiautomatic pistols and automatic guns, such as machine-guns.




1 First, the user draws the cocking handle back against the mainspring, compressing it. As the mainspring rebounds, it pushes the bolt forward, stripping a cartridge from the magazine and chambering it. The sear is connected to the trigger and now holds the bolt and the firing pin in position.

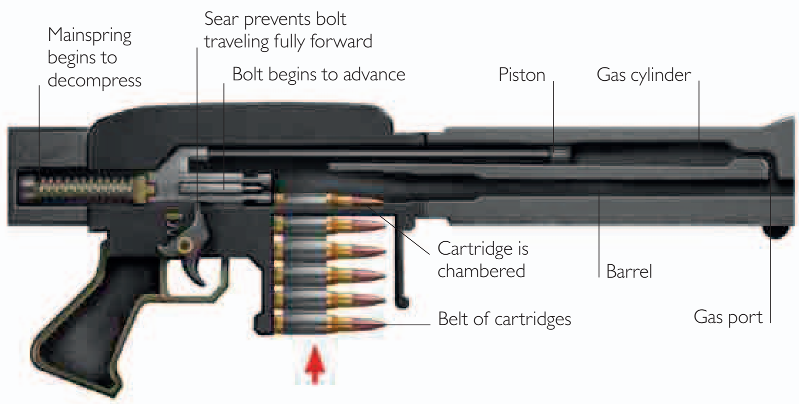



2 Pulling the trigger releases the sear. The mainspring extends fully, pushing the bolt fully forward and sending the firing pin flying toward the cartridge. The pin impacts the primer in the head of the cartridge and detonates it, igniting the propellant and firing the bullet.

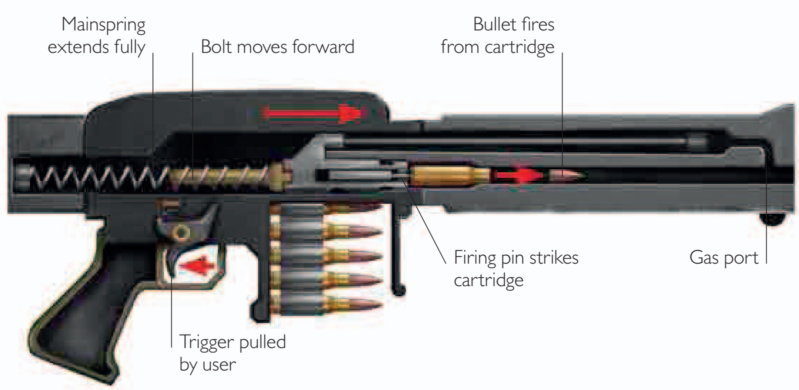

3 The recoil from firing the cartridge sends the bolt backwards, ejecting the empty cartridge case and allowing a new cartridge to enter the chamber. If the trigger remains depressed, the cycle continues.

## Gas reloading

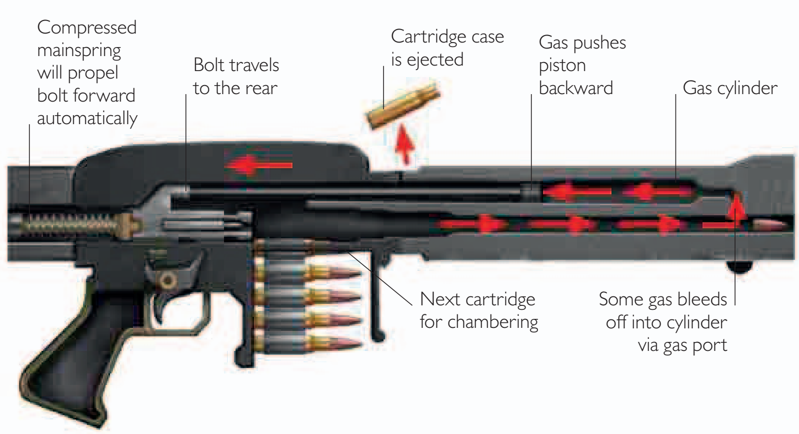
As an alternative to harnessing the force of the gun's recoil, it is possible to use some of the energy of the violently expanding gases that propel the bullet down the barrel. Some of that gas can be tapped off after the bullet has passed and employed to reload the gun by driving the breechblock or bolt to the rear. In automatic weapons, this action is cycled to produce continuous fire.




1 First, the user draws back the bolt against the mainspring. The mainspring pushes it forward again and as the bolt begins to advance, it strips a cartridge from the magazine and chambers it. The bolt is attached to a piston in a cylinder running parallel to the barrel. At the head of the cylinder is a gas port.




2 Pulling the trigger releases the sear. The mainspring extends, pushing the bolt forward. The firing pin impacts with the primer in the head of the cartridge, detonating it, igniting the propellant and firing the bullet.




3 As the bullet passes the gas port, some of the gas produced by burning the propellant bleeds through the port, forcing the piston backward. As the bolt travels to the rear, it ejects the spent cartridge case. The mainspring then extends, pushing the bolt ahead and chambering a new cartridge. If the trigger remains depressed, the cycle continues.